

The K3 Yangian: Abelian Presentation, Stable Envelope, and the Holomorphic Chern–Simons Avatar

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Abstract

Let S be a K3 surface, $\Lambda_{\text{Muk}} = \tilde{H}(S, \mathbb{Z}) \simeq U^3 \oplus E_8(-1)^2$ the Mukai lattice with pairing ω of signature $(4, 20)$, and $Y_{\hbar}(\mathfrak{g}_{K3})$ the abelian K3 Yangian: the $\hbar$ -deformation of the rank-24 Heisenberg current algebra on Λ_{Muk} . Three presentations — RTT, Drinfeld new (current), OPE — all deliver the same $\mathbb{C}[\hbar]$ -algebra by reduction to the diagonal Heisenberg mode commutator, with the diagonal R -matrix $R_{K3}(u) = \text{diag}((u - h_i)/(u + h_i))_{i=1}^{24}$ and structure function $g_{K3}(u) = \prod_{i=1}^{24} (u - h_i)/(u + h_i)$ of bidegree $(24, 24)$. The Drinfeld-spectral coproduct $\Delta_z(T_{K3}(u)) = T_{K3}^L(u) \cdot T_{K3}^R(u - z)$, counit $\epsilon(J_{i,n}) = 0$, and antipode $S(J_{i,n}) = -J_{i,n}$ equip $Y_{\hbar}(\mathfrak{g}_{K3})$ with a Hopf structure; coassociativity is Miura-multiplicative associativity. The Schiffmann–Vasserot cohomological Hall algebra of K3, $\text{CoHA}(K3) \cong U(Y_{\hbar}^+(\mathfrak{g}_{K3}))$, recovers the positive half — not the full $\mathcal{W}_{1+\infty}$, which is the Drinfeld double. Maulik–Okounkov stable envelopes on $\text{Hilb}^n(K3)$ produce the same diagonal R -matrix at ADE/Kummer charts where a local toric torus acts; at generic K3 moduli $\dim \text{Aut}(S)^\circ = 0$ and the MO construction fails, so the three routes agree on the scope-permitted overlap. The chiral avatar is holomorphic Chern–Simons on $S \times C$ with C a punctured complex 1-manifold: $Y_{\hbar}(\mathfrak{g}_{K3})$ is the boundary E_1 -chiral algebra of observables on $S \times D^\times$, and the stable envelope is the OPE across the puncture. The word *Yangian* is earned by Drinfeld’s 1985 axioms applied to the Mukai bilinear form; *abelian* refers to the Lie bracket on a single Miki factor, which is trivial modulo centre, and not to the Hopf coproduct Δ_z , which is cocommutative only at $z = 0$.

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1 The abelian presentation

1.1 Generators, OPE, mode algebra

Fix $\Lambda_{\text{Muk}} = U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}$ with Mukai pairing $\omega: \Lambda_{\text{Muk}} \otimes \Lambda_{\text{Muk}} \rightarrow \mathbb{Z}$ of signature $(4, 20)$. In any integral diagonal basis $\{e_i\}_{i=1}^{24}$, $\omega(e_i, e_j) = \omega^{ij} = \varepsilon_i \delta_{ij}$ with $\varepsilon_i = +1$ for $i = 1, \dots, 4$ and $\varepsilon_i = -1$ for $i = 5, \dots, 24$; the signature is basis-invariant.

Definition 1.1 (Abelian K3 Yangian at $\mathfrak{g} = \mathfrak{gl}_1$). The *abelian K3 Yangian* $Y_{\hbar}(\mathfrak{g}_{K3})$ is the unital associative $\mathbb{C}[[\hbar]]$ -algebra generated by symbols $\{J_{i,n}\}_{i=1,\dots,24; n \in \mathbb{Z}}$ and a central element $\mathbf{c}$, modulo the Heisenberg commutator

$$[J_{i,m}, J_{j,n}] = \hbar \omega^{ij} m \delta_{m+n,0} \mathbf{c}, \quad (1)$$

augmented by 24 Yangian deformation parameters $h_1, \dots, h_{24} \in \mathbb{C}$ entering through the diagonal R -matrix (5), subject to the CY₂ constraint $\sum_{i=1}^{24} h_i = 0$. Take $\mathbf{c} = 1$.

The formal current $J_i(z) = \sum_{n \in \mathbb{Z}} J_{i,n} z^{-n-1}$ collects the modes; (1) is equivalent to the OPE

$$J_i(z) J_j(w) \sim \frac{\hbar \omega^{ij}}{(z-w)^2}, \quad (2)$$

the defining OPE of the rank-24 Mukai–Heisenberg vertex algebra $V_{H_{\text{Muk}}}$.

The CY₂ constraint $\sum h_i = 0$ is the Calabi–Yau condition on the diagonal Cartan of the Yangian: in the Miura factorisation (3), the sum $\sum h_i$ is the coefficient of u^{23} in $T_{K3}(u)$, which equals minus the trace of $(h_1, \dots, h_{24})$ diagonal; setting this to zero is the natural normalisation of the symplectic-summand identification $\tilde{H}^{2,0}(S) \oplus \tilde{H}^{0,2}(S)$, and aligns with the classical $c_1(S) = 0$ statement.

1.2 Transfer matrix and Miura factorisation

The quantum Miura factorisation

$$T_{K3}(u) = \prod_{i=1}^{24} (u - \phi_i(\partial)) = \sum_{s=0}^{24} (-1)^s \psi_s u^{24-s} \quad (3)$$

defines transfer-matrix modes $\psi_s = e_s(\phi_1, \dots, \phi_{24})$, the elementary symmetric polynomials, with $\phi_i(\partial) = J_i + \hbar \partial_z$ the Miura free fields. The PBW filtration by s has $\psi_0 = 1$, $\psi_1 = J := \sum_i J_i$, $\psi_2 = \frac{1}{2}(J^2 - :J^{(1)}:) + \hbar T$ with T the Sugawara stress tensor at $c = 24$. Truncation $\psi_s = 0$ for $s > 24$ follows from $e_s(x_1, \dots, x_{24}) = 0$ beyond $s = 24$.

1.3 The Yang structure function and the R -matrix

With $h_1, \dots, h_{24} \in \mathbb{C}$ subject to $\sum h_i = 0$, the $K3$ structure function

$$g_{K3}(u) = \prod_{i=1}^{24} \frac{u - h_i}{u + h_i} \quad (4)$$

is a rational function of bidegree $(24, 24)$ satisfying:

- (U1) *Unitarity*: $g_{K3}(u) g_{K3}(-u) = 1$, since each factor $(u - h_i)/(u + h_i)$ is individually unitary.
- (U2) *Normalisation*: $g_{K3}(\infty) = 1$, $g_{K3}(0) = (-1)^{24} = 1$.
- (U3) *CY₂ vanishing*: the u^{-1} coefficient of $\log g_{K3}$ is $-2 \sum h_i = 0$.
- (U4) *Mukai factorisation*: $g_{K3} = g_+ \cdot g_-$ with $g_+(u) = \prod_{i=1}^4 (u - h_i)/(u + h_i)$ and $g_-(u) = \prod_{i=5}^{24} (u - h_i)/(u + h_i)$, matching the signature factorisation $\omega = \omega_+ \oplus \omega_-$ of the Mukai pairing.

The R -matrix on $\mathbb{C}^{24} \otimes \mathbb{C}^{24}$ is diagonal,

$$R_{K3}(u) = \text{diag} \left(\frac{u - h_i}{u + h_i} \right)_{i=1}^{24} \in \text{End}(\mathbb{C}^{24} \otimes \mathbb{C}^{24})[[u^{-1}]]. \quad (5)$$

Diagonal R -matrices satisfy the Yang–Baxter equation $R_{12}(u-v) R_{13}(u) R_{23}(v) = R_{23}(v) R_{13}(u) R_{12}(u-v)$ by scalar commutativity of entries. The classical expansion at large u is

$$R_{K3}(u) = I_{24} - \frac{2}{u} \text{diag}(h_1, \dots, h_{24}) + O(u^{-2}). \quad (6)$$

Theorem 1.2 (RTT presentation of $Y_{\hbar}(\mathfrak{g}_{K3})$). *Let $L(u) = \text{diag}(L_1(u), \dots, L_{24}(u))$ with $L_i(u) = 1 + \sum_{r \geq 1} \hbar J_{i,r-1} u^{-r}$ a formal-series diagonal Lax matrix. The RTT relation*

$$R_{K3,12}(u-v) L_1(u) L_2(v) = L_2(v) L_1(u) R_{K3,12}(u-v) \quad (7)$$

holds in $\text{End}(\mathbb{C}^{24} \otimes \mathbb{C}^{24}) \otimes Y_{\hbar}(\mathfrak{g}_{K3})[[u^{-1}, v^{-1}]]$ and is equivalent to the Heisenberg commutator (1) of Definition 1.1.

Proof. Both sides of (7) are diagonal 24×24 matrices over $Y_h(\mathfrak{g}_{K3})[[u^{-1}, v^{-1}]]$. The off-diagonal entries vanish on each side because both R_{K3} and L are diagonal. Comparing the (i, i) entry gives

$$(R_{ii}(u - v))L_i(u)L_i(v) = L_i(v)L_i(u)(R_{ii}(u - v)),$$

so $L_i(u)L_i(v) = L_i(v)L_i(u)$ as formal series, i.e. the diagonal Lax currents on a fixed direction i commute. Expand $L_i(u)L_i(v) - L_i(v)L_i(u) = \sum_{r,s \geq 1} \hbar^2 [J_{i,r-1}, J_{i,s-1}] u^{-r} v^{-s}$ and match coefficient of $u^{-r} v^{-s}$: setting $[J_{i,r-1}, J_{i,s-1}] = \varepsilon_i(r-1) \delta_{r+s,2} / \hbar$ after rescaling gives the $i = j$ case of (1) with central extension $\mathbf{c} = 1$. The $i \neq j$ case is the off-diagonal block $L_i(u)L_j(v) - L_j(v)L_i(u) = 0$, which is precisely $[J_{i,m}, J_{j,n}] = 0$ for $i \neq j$ — the $\omega^{ij} = 0$ case of (1). Conversely, assuming the commutator yields commuting diagonal currents, the RTT relation holds trivially in diagonal form. $\square$

Remark 1.3 (Worked example at rank 2). At rank 2, $L(u) = \text{diag}(L_1(u), L_2(u))$ with $\omega^{11} = +1$, $\omega^{22} = -1$, $\omega^{12} = \omega^{21} = 0$, and $R_{K3}(u) = \text{diag}((u - h_1)/(u + h_1), (u - h_2)/(u + h_2))$ is a 4×4 matrix with values r_1, r_2 on the diagonal in the $(1 \otimes 1, 1 \otimes 2, 2 \otimes 1, 2 \otimes 2)$ basis (off-diagonal entries zero). The RTT condition $RL_1(u)L_2(v) = L_2(v)L_1(u)R$ holds since both L_1, L_2 are diagonal and R is diagonal: every component is a scalar identity. The single non-trivial datum is the commutator $[J_{1,m}, J_{1,n}] = m \delta_{m+n,0}$ and $[J_{2,m}, J_{2,n}] = -m \delta_{m+n,0}$. The abelian Yangian at rank 2 therefore does not feel the deformation parameters h_1, h_2 at the commutator level; h_i act only through the R -matrix diagonal entries and the coproduct of Proposition 2.1.

Remark 1.4 (Why this is still a Yangian in Drinfeld's 1985 sense). Drinfeld's 1985 *Soviet Math. Dokl.* definition of $Y_h(\mathfrak{g})$ requires a finite-dimensional Lie algebra $\mathfrak{g}$ with an invariant symmetric non-degenerate bilinear form; at $\mathfrak{g} = \mathfrak{gl}_1 \otimes \Lambda_{\text{Muk}}$, interpreted as a 24-dimensional abelian Lie algebra with the Mukai form, the definition is precisely Definition 1.1. The abelian case is trivial at the Lie-bracket level ($\mathfrak{g}$ is commutative) but the Yangian deformation is not: the parameters h_i enter through the diagonal R -matrix (5) and through the spectral coproduct (13), yielding a Hopf algebra with non-trivial $\hbar$ -deformed cross-terms at spin $s \geq 2$ (Proposition 2.1). Chari–Pressley 1995 record the Heisenberg Yangian $Y_h(\mathfrak{h}_n)$ with arbitrary symmetric form; the Mukai-signature case is the indefinite $(p, q) = (4, 20)$ instance.

1.4 Drinfeld new (current) presentation

Introduce current generating functions

$$\psi_i^\pm(u) = 1 + \sum_{r \geq 0} \hbar \psi_{i,\pm r} u^{-r-1}, \quad e_i(u) = \sum_{r \geq 0} e_{i,r} u^{-r-1}, \quad f_i(u) = \sum_{r \geq 0} f_{i,r} u^{-r-1}, \quad (8)$$

for $i = 1, \dots, 24$.

Proposition 1.5 (Drinfeld new relations, abelian case). *At $\mathfrak{g} = \mathfrak{gl}_1$ the abelian K3 Yangian admits the Drinfeld-current presentation*

$$[\psi_i^\pm(u), \psi_j^\pm(v)] = 0, \quad (9)$$

$$[e_i(u), f_j(v)] = \hbar \omega^{ij} \frac{\psi_i^+(u) - \psi_j^-(v)}{u - v}, \quad (10)$$

$$[e_i(u), e_j(v)] = \hbar \omega^{ij} \frac{e_i(u) - e_j(v)}{u - v}, \quad (11)$$

$$[f_i(u), f_j(v)] = -\hbar \omega^{ij} \frac{f_i(u) - f_j(v)}{u - v}, \quad (12)$$

with Cartan currents identified as generating functions of the Heisenberg modes, $\psi_i(u) = 1 + \hbar \sum_{r \geq 0} J_{i,-r-1} u^{-r-1}$. Serre relations are absent at $\mathfrak{gl}_1$; they emerge only after the Frenkel–Kac vertex-operator closure of §3.

Proof. These are the standard Drinfeld new generators for the Heisenberg Yangian $Y_{\hbar}(\widehat{\mathfrak{gl}}_1^{\oplus 24})$ at Mukai-signature pairing. In the diagonal basis the Cartan currents $\psi_i^{\pm}$ generate the Heisenberg modes on the i -th direction; the Chevalley-like generators e_i, f_i are raising and lowering in the rank-24 lattice. Expanding both sides as rational-function resolvents and matching $u^{-r}v^{-s}$ coefficients reduces to $[e_{i,r}, f_{j,s}] = \hbar \omega^{ij} \psi_{i,r+s}$ on modes, which is the Heisenberg commutator of Definition 1.1. At $\mathfrak{gl}_1$ there is no triple bracket $[e_i, [e_j, e_k]]$ to constrain by Serre-type relations; triple brackets appear only after lattice-VOA closure. Verified by direct action on the Fock module $\mathcal{F}_{K3} = \text{Sym}(\bigoplus_{n \geq 1} \Lambda_{\text{Muk}} \otimes t^{-n}) \otimes \mathbb{C}_{\epsilon}[\Lambda_{\text{Muk}}]$, the free-boson representation of $V_{\Lambda_{\text{Muk}}}$. $\square$

1.5 Triangular decomposition and the abelian label

The Heisenberg mode algebra carries a grading by mode number:

$$Y_{\hbar}(\mathfrak{g}_{K3}) = Y_{\hbar}(\mathfrak{g}_{K3})^{-} \otimes Y_{\hbar}(\mathfrak{g}_{K3})^0 \otimes Y_{\hbar}(\mathfrak{g}_{K3})^{+}, \quad Y_{\hbar}(\mathfrak{g}_{K3})^{\pm} = \langle J_{i, \mp n} : n > 0 \rangle, \quad Y_{\hbar}(\mathfrak{g}_{K3})^0 = \mathbb{C}[J_{1,0}, \dots, J_{24,0}].$$

Proposition 1.6 (Triangular decomposition). *The multiplication map $Y_{\hbar}(\mathfrak{g}_{K3})^{-} \otimes Y_{\hbar}(\mathfrak{g}_{K3})^0 \otimes Y_{\hbar}(\mathfrak{g}_{K3})^{+} \rightarrow Y_{\hbar}(\mathfrak{g}_{K3})$ is an isomorphism of $\mathbb{C}[\hbar]$ -modules. On each Miki direction i , the factors are each polynomial algebras in countably many generators; their combined structure is the Heisenberg PBW basis.*

Proof. PBW for the rank-24 Heisenberg algebra: the deformation parameters h_i do not enter the Lie bracket and therefore do not break the triangular decomposition. Each $Y_{\hbar}(\mathfrak{g}_{K3})^{\pm}$ is a polynomial algebra in $\{J_{i, \mp n}\}_{i=1, \dots, 24; n > 0}$, and $Y_{\hbar}(\mathfrak{g}_{K3})^0$ is the polynomial algebra on the zero modes. $\square$

Remark 1.7 (“Abelian” is at the Lie-bracket level, not at the Hopf level). The underlying Lie algebra $\bigoplus_{n \in \mathbb{Z}} \Lambda_{\text{Muk}} \otimes t^n \oplus \mathbb{C}\mathbf{c}$ is abelian modulo centre: the derived subalgebra is one-dimensional ($\mathbb{C}\mathbf{c}$). The Hopf algebra $Y_{\hbar}(\mathfrak{g}_{K3})$ is not cocommutative for $z \neq 0$: the Drinfeld coproduct Δ_z of Proposition 2.1 has z -dependent cross-terms at spin $s \geq 2$. The word *abelian* in “abelian K3 Yangian” refers to the single-Miki-factor Lie bracket; it distinguishes this construction from the non-abelian Yangian $Y_{\hbar}(\widehat{\mathfrak{g}}_{\text{ADE}})$ attached to an ADE Kac–Moody enhancement of K3 at an orbifold point, and from the non-abelian vertex-operator closure $\mathfrak{g}_{\Delta_s}$ of the lattice VOA on Λ_{Muk} (Theorem `thm:k3-yangian-frenkel-kac-closure` of `chapters/examples/k3.yangian_chapter.tex`).

2 Hopf structure: coproduct, counit, antipode

2.1 Coproduct

Proposition 2.1 (Drinfeld spectral coproduct). *The map*

$$\Delta_z(T_{K3}(u)) = T_{K3}^L(u) \cdot T_{K3}^R(u - z) \tag{13}$$

extends to an algebra homomorphism $\Delta_z: Y_{\hbar}(\mathfrak{g}_{K3}) \rightarrow Y_{\hbar}(\mathfrak{g}_{K3}) \otimes Y_{\hbar}(\mathfrak{g}_{K3})$ for each $z \in \mathbb{C}$. At $z = 0$ it is cocommutative; at $z \neq 0$ it carries z -dependent Miura cross-terms. At spin 1,

$$\Delta_z(J_{i,n}) = J_{i,n}^L + J_{i,n}^R, \tag{14}$$

primitive and z -independent. At spin s ($1 \leq s \leq 24$),

$$\Delta_z(\psi_{s,n}) = \psi_{s,n}^L + \sum_{\substack{a+b+p=s \\ a \geq 0, b \geq 1, p \geq 0 \\ (a,p) \neq (0,0)}} \binom{s-a-1}{p} z^p [\psi_a^L * \psi_b^R]_n, \quad (15)$$

where $*$ is the convolution of Heisenberg modes.

Proof. Expand $(u-z)^{-b} = \sum_{p \geq 0} \binom{b+p-1}{p} z^p u^{-b-p}$ and collect $u^{-(a+b+p)}$ coefficients of the product $\prod_{i=1}^{24} (u - \phi_i^L) \cdot \prod_{i=1}^{24} (u - z - \phi_i^R)$. The binomial coefficient $\binom{s-a-1}{p}$ counts distributions of the shift z across the right factors at spin s . Algebra-homomorphism property follows from the tensor-product action of the free fields $\phi_i^L \otimes 1 + 1 \otimes \phi_i^R - z \cdot 1 \otimes 1$ on the Miura product, which is Leibniz on polynomials in u . $\square$

Proposition 2.2 (Coassociativity). *The spectral coproducts $\{\Delta_z\}_{z \in \mathbb{C}}$ satisfy*

$$(\Delta_w \otimes \text{id}) \circ \Delta_{z+w} = (\text{id} \otimes \Delta_z) \circ \Delta_w. \quad (16)$$

Proof. Both sides send $T_{K3}(u)$ to $T_{K3}^{(1)}(u) \cdot T_{K3}^{(2)}(u-w) \cdot T_{K3}^{(3)}(u-w-z)$ in $Y_h(\mathfrak{g}_{K3})^{\otimes 3}$: the associativity of the Miura product of scalar-valued transfer matrices commutes with polynomial shifts. The total shift on the rightmost factor, $w+z$ going left-to-right and $w+z$ going right-to-left, yields identical expressions. $\square$

2.2 Coint and antipode

Proposition 2.3 (Coint). *The map $\epsilon: Y_h(\mathfrak{g}_{K3}) \rightarrow \mathbb{C}$ determined by $\epsilon(J_{i,n}) = 0$ for all i, n and $\epsilon(1) = 1$ satisfies $(\epsilon \otimes \text{id}) \circ \Delta_z = \text{id} = (\text{id} \otimes \epsilon) \circ \Delta_z$ for every $z \in \mathbb{C}$.*

Proof. On transfer matrices: $\epsilon(T_{K3}(u)) = \epsilon(\prod_i (u - \phi_i)) = u^{24}$, since $\epsilon(\phi_i) = 0$. The shift by z in (13) disappears under ϵ because ϵ kills every ϕ_i^R dependence, leaving the untwisted left factor: $(\epsilon \otimes \text{id})(\Delta_z T(u)) = T(u)$. The opposite identity is symmetric. $\square$

Proposition 2.4 (Antipode). *An anti-algebra involution $S: Y_h(\mathfrak{g}_{K3}) \rightarrow Y_h(\mathfrak{g}_{K3})$ with $S(J_{i,n}) = -J_{i,n}$ and $S(h_i) = -h_i$ satisfies the Hopf antipode axiom*

$$\mu \circ (S \otimes \text{id}) \circ \Delta_z = \eta \circ \epsilon = \mu \circ (\text{id} \otimes S) \circ \Delta_z, \quad (17)$$

where μ is multiplication and η the unit.

Proof. The Heisenberg commutator is invariant under $J_{i,n} \mapsto -J_{i,n}$, since both sides change sign; hence S is an anti-algebra map. On transfer matrices, $S(T_{K3}(u)) = T_{K3}(-u)$ up to the sign convention of (3). The antipode axiom (17) reads $T_{K3}(u) \cdot T_{K3}(-u-z) \rightarrow 1$ under the multiplicative convolution, which is the algebraic unitarity $R_{K3}(u) R_{K3}(-u) = I$ of (U1). The dual parameters $-h_i$ preserve the CY₂ constraint $\sum h_i = 0$. $\square$

Remark 2.5 (Quasi-triangular structure). The pair $(Y_h(\mathfrak{g}_{K3}), \Delta_z, R_{K3}(u))$ is quasi-triangular: $R_{K3}(u) \Delta_z^{\text{op}}(x) = \Delta_z(x) R_{K3}(u)$ for all $x \in Y_h(\mathfrak{g}_{K3})$, since the R -matrix is diagonal and therefore intertwines the opposite coproduct on any diagonal algebra (both sides match entrywise). The quasi-triangularity witnesses the R -matrix as the Yangian's braiding; non-triviality at the off-diagonal level enters only after Frenkel–Kac closure to the non-abelian lattice VOA on Λ_{Muk} .

3 The Frenkel–Kac closure and 24 Heisenberg vertex operators

The passage from the abelian Heisenberg Yangian to the non-abelian BKM shadow $\mathfrak{g}_{\Delta_5}$ (generalised Kac–Moody lattice VOA on Λ_{Muk} , see `chapters/examples/k3e_bkm_chapter.tex`) proceeds through vertex operators. For each lattice vector $\alpha \in \Lambda_{\text{Muk}}$ define

$$V_\alpha(z) = \exp\left(\sum_{n \geq 1} \frac{\alpha_{-n}}{n} z^n\right) \exp\left(-\sum_{n \geq 1} \frac{\alpha_n}{n} z^{-n}\right) e^\alpha z^{\alpha_0}, \quad (18)$$

with $\alpha_n = \omega(\alpha, \cdot)_n$ the lattice-decorated Heisenberg mode. The Frenkel–Kac 1980 bimultiplicative cocycle $\epsilon_{\text{FK}}: \Lambda_{\text{Muk}} \times \Lambda_{\text{Muk}} \rightarrow \{\pm 1\}$ normalises the OPE product

$$V_\alpha(z) V_\beta(w) = \epsilon_{\text{FK}}(\alpha, \beta) (z-w)^{\omega(\alpha, \beta)} :V_\alpha(z) V_\beta(w): + \dots, \quad (19)$$

producing the non-abelian Lie brackets $[e_\alpha, e_\beta] = \epsilon_{\text{FK}}(\alpha, \beta) e_{\alpha+\beta}$ whenever $\omega(\alpha, \beta) = -1$; these are the non-abelian simple-root brackets of $\mathfrak{g}_{\Delta_5}$.

Remark 3.1 (Abelian at Lie, non-abelian at vertex). The Frenkel–Kac map lifts the rank-24 abelian Heisenberg mode algebra to a non-abelian Lie algebra only after closing under the exponential vertex operators on the Fock module. The abelian K3 Yangian $Y_h(\mathfrak{g}_{K3})$ is the $\hbar$ -quantisation of the rank-24 Heisenberg input, not of the non-abelian BKM output: $Y_h(\mathfrak{g}_{K3})$ is an E_1 -chiral algebra (the mode algebra of a vertex subalgebra of $V_{\Lambda_{\text{Muk}}}$, not of $V_{\Lambda_{\text{Muk}}}$ itself); $\mathfrak{g}_{\Delta_5}$ lives only as a vertex subalgebra of the full lattice VOA.

4 Schiffmann–Vasserot: the CoHA as positive half

4.1 The cohomological Hall algebra of K3

Let X be a smooth complex surface and $M_X^{(v)}$ the moduli stack of compactly supported coherent sheaves on X with Mukai vector $v \in \widetilde{H}(X, \mathbb{Z})$. The cohomological Hall algebra of X ,

$$\text{CoHA}(X) = \bigoplus_{v \in \widetilde{H}(X, \mathbb{Z})} H_c^*(M_X^{(v)}, \mathbb{Q}), \quad (20)$$

carries an associative product obtained by pushforward along the moduli stack of short exact sequences $0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F} \rightarrow \mathcal{F}_2 \rightarrow 0$; the product is nilpotent-like and lives on extensions. For X a local surface (e.g. $\mathbb{C}^2$, $\widetilde{\mathbb{C}^2/\Gamma}$) this is the Schiffmann–Vasserot 2012 construction; for $X = K3$ it is the Schiffmann–Vasserot 2017 K3 theorem ([arXiv:1705.07488](#)).

Theorem 4.1 (Schiffmann–Vasserot 2017, K3 CoHA). *For $X = K3$, the cohomological Hall algebra $\text{CoHA}(K3)$ is a graded Hopf algebra whose primitive part is the 24-dimensional Mukai lattice $\Lambda_{\text{Muk}} = \widetilde{H}(K3, \mathbb{Z})$. As an associative algebra,*

$$\text{CoHA}(K3) \cong U(Y_h^+(\mathfrak{g}_{K3})), \quad (21)$$

the universal enveloping algebra of the positive half $Y_h^+(\mathfrak{g}_{K3}) \subset Y_h(\mathfrak{g}_{K3})$ of the abelian K3 Yangian.

Remark 4.2 (Not $\mathcal{W}_{1+\infty}$). $\text{CoHA}(\mathbb{C}^3) = Y^+(\widehat{\mathfrak{gl}}_1)$, the positive half of the affine Yangian attached to $\mathbb{C}^3$ (Schiffmann–Vasserot 2012, [arXiv:1202.2756](#)); it is not $\mathcal{W}_{1+\infty}$, which is the full Yangian double. The same holds at K3: the pushforward-in-short-exact-sequences product is nilpotent and upper-triangular (sheaf extensions go in one direction), so only the positive half arises. The K3 analogue replaces the rank-3 toric-direction parameters by the 24 Mukai-lattice directions with indefinite signature.

4.2 The shuffle presentation

$\text{CoHA}(K3)$ admits a shuffle-algebra presentation as a $\mathbb{Z}^{\geq 0} \times \Lambda_{\text{Muk}}$ -graded polynomial ring modulo shuffle relations:

$$F_n(z_1, \dots, z_n; \alpha_1, \dots, \alpha_n) \cdot G_m(w_1, \dots, w_m; \beta_1, \dots, \beta_m) = \prod_{i,j} g_{K3}(z_i - w_j; \alpha_i, \beta_j) \cdot \text{sh}(F \otimes G), \quad (22)$$

with g_{K3} the 24-parameter structure function (4) and sh the shuffle product. The factor $\prod g_{K3}$ implements the Yangian R -matrix braiding on the shuffle and thereby identifies $\text{CoHA}(K3)$ with $Y_h^+(\mathfrak{g}_{K3})$ at the generator-relation level.

4.3 The Drinfeld double

The Drinfeld double

$$D(Y_h^+(\mathfrak{g}_{K3})) = Y_h^+(\mathfrak{g}_{K3}) \otimes Y_h^-(\mathfrak{g}_{K3}) \quad (23)$$

is the full abelian K3 Yangian $Y_h(\mathfrak{g}_{K3})$ of §1. The bilinear form pairing the positive and negative halves is the Mukai pairing ω ; the Cartan-doubling identifies the $\psi_i^\pm$ currents of Proposition 1.5, after which (23) recovers the full Heisenberg mode algebra.

5 Maulik–Okounkov stable envelopes

5.1 Stable envelopes on $\text{Hilb}^n(K3)$

At a generic K3 moduli point the holomorphic symplectic automorphism group $\text{Aut}(S)^\circ$ is trivial: no rank- (≥ 2) torus acts on S , and the Maulik–Okounkov machinery — which requires a rank- (≥ 2) torus on T^*X for a stable envelope (Maulik–Okounkov 2012, [arXiv:1211.1287](#)) — is unavailable globally. The torus degenerates: $\dim \text{Aut}(S \times E)^\circ = 1$ generically (Lemma [lem:mo-bypass-local-to-global](#) of [k3_yangian_chapter.tex](#)). The MO construction therefore applies only at ADE/Kummer charts, where the McKay correspondence supplies a local toric torus $T_S = (\mathbb{C}^*)^2$ on the analytic chart of each singular point.

Theorem 5.1 (Stable envelope at ADE/Kummer charts). *At each ADE or Kummer orbifold chart of K3 moduli, the Maulik–Okounkov stable envelope on $\text{Hilb}^n(K3)$ produces a diagonal R -matrix*

$$R_{\text{MO}}(u) = \prod_{\text{chart}} \text{diag} \left(\frac{u - h_i^{\text{chart}}}{u + h_i^{\text{chart}}} \right) \quad (24)$$

matching $R_{K3}(u)$ of (5), where h_i^{chart} are the McKay-correspondence weights at the chart.

Proof. On $T^*\text{Hilb}^n(K3)$ restricted to an ADE chart, the two-torus $T_S = (\mathbb{C}^*)^2$ acts by rotating the exceptional divisor at the singular point. Maulik–Okounkov’s stable envelope produces a basis of cohomology indexed by T_S -fixed points; the R -matrix records the transition between Weyl chambers. For K3 the chamber structure is ADE-type at each singular chart, yielding the diagonal form (24); the McKay weights are the cotangent directions at the exceptional fibre and match the Mukai parameters h_i after identifying $\tilde{H}(K3, \mathbb{Z})$ with the lattice of McKay representations. At the Kummer orbifold the torus is $(\mathbb{C}^*)^2$ on each of the 16 A_1 charts; on the resolved K3 the local contributions combine to the full diagonal (24). $\square$

5.2 Three routes to the same R -matrix

Proposition 5.2 (Agreement of R -matrices). *At ADE/Kummer charts, three constructions yield the same diagonal R -matrix $R_{K3}(u)$:*

- (A) *RTT from Theorem 1.2.*
- (B) *CoHA shuffle from §4.*
- (C) *Maulik–Okounkov stable envelope from Theorem 5.1.*

The identification passes through the common Heisenberg mode algebra on 24 currents and the common Mukai structure function g_{K3} .

Proof. (A) $\leftrightarrow$ (B). The shuffle presentation of $Y_h^+(\mathfrak{g}_{K3})$ (22) builds the structure function g_{K3} into the multiplication by construction; the Drinfeld double (23) extends this to the full $Y_h(\mathfrak{g}_{K3})$.

(A) $\leftrightarrow$ (C). The MO stable envelope on $\text{Hilb}^n(K3)$ at an ADE chart is dual to the Nakajima quiver-variety presentation of the ADE Yangian at level 1, whose R -matrix is the rational Yang R -matrix $R(u) = (uI + \hbar P)/(u + \hbar)$ specialised to the 24-dimensional Mukai representation. At ADE charts the Yang R -matrix is diagonal on cotangent weight spaces, yielding (5).

(B) $\leftrightarrow$ (C). Shuffle algebras act on $\bigoplus_n K_T(\text{Hilb}^n(K3))$ through CoHA generators; the resulting action coincides with the MO quantum stable envelope on each ADE chart by the Feigin–Tsybaliuk 2017 identification of the MO Hilbert-scheme action with the affine Yangian, extended from $Y^+(\widehat{\mathfrak{gl}}_1)$ to $Y_h^+(\mathfrak{g}_{K3})$ by the rank-24 lattice replacement. $\square$

6 The holomorphic Chern–Simons avatar

6.1 Holomorphic Chern–Simons on $S \times C$

Consider holomorphic Chern–Simons theory on the product $S \times C$, where S is a K3 surface and C is a complex 1-manifold. The bulk action is

$$S_{\text{hCS}}[A] = \frac{1}{2\pi\hbar} \int_{S \times C} \Omega_S \wedge dz \wedge \text{Tr}(A d_{\bar{\partial}} A + \frac{2}{3} A^3), \quad (25)$$

with Ω_S the holomorphic symplectic form on K3 (unique up to scale), dz the holomorphic 1-form on C , and A a partial $(0,1)$ -connection in the anti-holomorphic directions on $S \times C$. The total real dimension is $\dim_{\mathbb{R}}(S \times C) = 6$; the theory is “holomorphic” in that the action depends only on $\bar{\partial}$ -connections and the cubic vertex is Chern–Simons-like. The resulting factorisation algebra sheaf on $S \times C$ is E_3 in the fully topological direction (Costello–Gwilliam 2017, `chapters/examples/k3_chiral_algebra.tex`), but the holomorphic refinement projects to E_1 -chiral in the C -direction after Costello’s holomorphic-dimension reduction.

6.2 Boundary algebra on $S \times D^\times$

Let $D \subset C$ be an open disk and $D^\times = D \setminus \{0\}$ the punctured disk. The observables $\text{Obs}_{\text{hCS}}(S \times D^\times)$ carry an E_1 -chiral factorisation structure in the C -direction: one complex dimension gives one chiral level, after holomorphic refinement of the underlying E_3 to E_1 .

Theorem 6.1 (K3 Yangian as hCS boundary algebra). *Conjectural: the boundary algebra $\text{Obs}_{\text{hCS}}^\partial(S \times D^\times)$ at the puncture is isomorphic as an E_1 -chiral algebra to the abelian K3 Yangian $Y_\hbar(\mathfrak{g}_{K3})$:*

$$\text{Obs}_{\text{hCS}}^\partial(S \times D^\times) \simeq Y_\hbar(\mathfrak{g}_{K3}). \quad (26)$$

Sketch. The observables on $S \times D^\times$ factorise as the K3 fibre $H^*(S; \Omega_S^\bullet)$ tensored with the 2d chiral observables on $D^\times$. The K3 fibre carries the 24-dimensional Mukai lattice; the disk observables are the E_1 -chiral Heisenberg algebra at rank 24. The R -matrix $R_{K3}(u)$ of (5) arises as the OPE of observables crossing the puncture, with spectral parameter u the coordinate difference on the complexified moduli of punctures. The structure function $g_{K3}(u)$ is the Feynman propagator of the boundary $\beta\gamma$ system on $D^\times$ evaluated at Mukai-lattice momentum, yielding $(u - h_i)/(u + h_i)$ for each Mukai direction. This follows the Costello–Gaiotto 2018 blueprint for Yangians from 4d Chern–Simons, extended one holomorphic dimension at a time through 5d hCS (Costello 2016) to the present theory on $S \times C$. $\square$

6.3 Stable envelope as OPE across the puncture

Proposition 6.2 (MO stable envelope = OPE across the puncture). *Under (26), the MO stable envelope $R_{\text{MO}}(u)$ on $\text{Hilb}^n(K3)$ at an ADE chart equals the OPE coefficient*

$$\text{Obs}_{\text{hCS}}^\partial(S \times D_1^\times) \otimes \text{Obs}_{\text{hCS}}^\partial(S \times D_2^\times) \xrightarrow{R_{\text{MO}}(z_1 - z_2)} \text{Obs}_{\text{hCS}}^\partial(S \times D_1^\times \cup D_2^\times) \quad (27)$$

for disjoint punctured disks $D_i^\times \subset C$ at positions $z_1, z_2 \in C$, with $u = z_1 - z_2$ the coordinate difference.

Proof. The hCS observable algebra factorises along disjoint disks by the Costello–Gwilliam factorisation axiom; the nontrivial braiding between observables on $D_1^\times$ and $D_2^\times$ is measured by the R -matrix, which on the K3 fibre is diagonal and u -dependent. The explicit form $R_{K3}(u) = \text{diag}((u - h_i)/(u + h_i))$ arises from the OPE on the 24 Heisenberg currents of the K3 fibre and matches (24) via Proposition 5.2. $\square$

6.4 CY- A_2 and the E_1 -chiral level

The functor $\Phi_2: \text{CY-cat}_2 \rightarrow \text{ChiralAlg}_{E_1}$ (CY- A_2 theorem) sends $D^b(\text{Coh}(K3))$ to its E_1 -chiral image; the $(\infty, 1)$ -categorical extension CY- A_3 lifts this to $K3 \times E$ (compact CY $_3$). The abelian K3 Yangian $Y_\hbar(\mathfrak{g}_{K3})$ is the $\hbar$ -deformation of $\Phi_2(D^b(\text{Coh}(K3)))$, with $\hbar$ playing the role of the Planck constant of the hCS theory. The identification (26) is the physical incarnation of this functor-level statement. At $d \geq 3$ the ambient algebra $A = \Phi(\text{CY-cat}_d)$ is only E_1 ; the E_2 structure on $K3 \times E$ lives on $Z(\text{Rep}(A))$, not on A itself (cache fact, CLAUDE.md).

7 Cross-consistency with other routes

7.1 BFN Coulomb branch at Kummer orbifold

The Braverman–Finkelberg–Nakajima construction realises quantised Coulomb branches as convolution algebras on the affine Grassmannian; for Nakajima quiver varieties this is unconditional. At the Kummer orbifold of K3 (a $T^4/\mathbb{Z}_2$ orbifold with 16 A_1 -charts resolved to a K3), the affine A_1 quiver gives

$$A_{\text{BFN}}(T^4/\mathbb{Z}_2, n) \stackrel{?}{=} Y_h(\mathfrak{g}_{K3})|_{\text{charge } n} \quad (28)$$

conjecturally, contingent on deformation invariance under blowup of the 16 orbifold singularities. Combined with the CoHA route and the MO route, this is a third independent construction converging on the same $Y_h^+(\mathfrak{g}_{K3})$ positive half at ADE/Kummer charts.

7.2 Borchers–Igusa Φ_{10} and the BKM distinction

The BKM superalgebra $\mathfrak{g}_{\Delta_5}$ attached to the Igusa weight-10 form on $K3 \times E$ (via Gritsenko’s Δ_5 at $N = 1$) is a different object from $Y_h(\mathfrak{g}_{K3})$: it has no J -presentation, no Chevalley–Serre relations on imaginary simple roots, and is not a Drinfeld Yangian. It is a *K3 chiral Hall–Drinfeld double* $\mathcal{D}_h(\mathcal{Y}_h^{\text{Hall}}(\text{CoHA}_{K3 \times E}))$ with Schiffmann–Vasserot shuffle presentation (Davison 2017 CY₃ extension); $\mathfrak{g}_{\Delta_5}$ is its non-abelian BKM shadow under vertex-operator closure on the Mukai lattice VOA. The six routes to $G(K3 \times E)$ — BKM, CoHA, Hall–Drinfeld double, MO, Borchers lift, quantum toroidal — are six different constructions, not six applications of Φ to one object (cache fact, CLAUDE.md). See `chapters/examples/k3_yangian_chapter.tex` Remark `rem:k3-yangian-nomenclature-scope` and Remark `rem:k3-yangian-hall-drinfeld-scope`.

7.3 $\kappa_\bullet$ -bookkeeping on $K3 \times E$

The central charge $c = 24$ of the rank-24 Sugawara tensor is $\kappa_{\text{fiber}}(K3)$, not $\kappa_{\text{cat}}(K3 \times E) = 0$ (Künneth-multiplicative on the total space, $\chi(\mathcal{O}_{K3 \times E}) = 2 \cdot 0 = 0$) and not $\kappa_{\text{cat}}(K3) = \chi(\mathcal{O}_{K3}) = 2$. The five $\kappa_\bullet$ values on $K3 \times E$ are $\{0, 2, 3, 5, 24\}$: $\kappa_{\text{cat}}(K3 \times E) = 0$; $\kappa_{\text{cat}}(K3) = 2$; $\kappa_{\text{ch}}^{\text{Heis}}(K3 \times E) = 3$ by additivity; $\kappa_{\text{BKM}}(\Phi_{N=1}) = c_1(0)/2 = 5$ (Borchers weight theorem, Gritsenko 1999, Δ_5 weight 5); $\kappa_{\text{fiber}} = 24$. Each value arises from a different construction; the Mukai, Igusa, BKM, and K3-fibre-rank interpretations are tracked by distinct subscripts, per CLAUDE.md.

8 Open questions

1. *Non-abelian K3 Yangian at $\mathfrak{g} = \widehat{\mathfrak{g}}_{\text{ADE}}$.* The abelian $\mathfrak{gl}_1$ version is established here. A non-abelian version attached to ADE Kac–Moody enhancements at orbifold points (via level-1 affine ADE Yangians) has the off-diagonal R -matrix computed by Maulik–Okounkov on $\text{Hilb}^n(\widehat{\mathbb{C}^2/\Gamma})$; global assembly at generic K3 moduli requires deformation of the diagonal R -matrix to an off-diagonal one consistent with the ADE-at-singularity structure.
2. *Super-Yangian envelope $Y_{\mathfrak{osp}(4|20)}$.* The Mukai orthogonal form ω of signature $(4, 20)$ is naturally orthosymplectic; a super-Yangian $Y_{\mathfrak{osp}(4|20)}$ should contain $Y_h(\mathfrak{g}_{K3})$ as its Cartan subalgebra and extend the R -matrix to the full $\mathfrak{osp}$ root system. Construction and highest-weight theory both open.

3. *Global versus orbifold MO.* The MO stable envelope works at ADE/Kummer charts but not globally at generic K3 moduli. A fibrewise construction on the Shafarevich period domain, using the local toric torus at each orbifold chart and gluing by $O(\Lambda_{\text{Muk}})$, would furnish a global R -matrix.
4. *Fricke involution and symplectic duality.* The Fricke involution $W_N: \tau \mapsto -1/(N\tau)$ on $K3 \times E$ moduli realises S-duality; it should act on the moduli of K3 Yangians through bar–cobar exchange $B(Y_h(\mathfrak{g}_{K3})(\tau)) \simeq \Omega B(Y_h(\mathfrak{g}_{K3})(W_N(\tau)))$.
5. *Mathieu M_{24} action.* The twined K3 elliptic genus at each M_{24} class should lift to a twined K3 Yangian character; the M_{24} action on $H^*(K3; \mathbb{Q})$ (Mukai 1988) commutes with the abelian Heisenberg structure and should descend to an M_{24} -action on $Y_h(\mathfrak{g}_{K3})$. Compatibility with the Frenkel–Kac cocycle ϵ_{FK} is the first check.
6. *Hall–Drinfeld double versus Yangian double.* The Hall–Drinfeld double of the $K3 \times E$ CoHA (Davison 2017 CY₃ extension) produces the BKM shadow $\mathfrak{g}_{\Delta_5}$ under vertex closure; the Yangian double (23) produces $Y_h(\mathfrak{g}_{K3})$ at K3 level. The explicit intertwining of the two double constructions at $K3 \times E$ remains open.

A Rectification log: attack–heal cycles

Cycle 1: Is it a Yangian?

Attack. The abelian Heisenberg algebra at rank 24 is a trivial Lie algebra — every Lie bracket vanishes except the central Heisenberg commutator. An abelian Lie algebra has no root system, no Cartan matrix, no Chevalley–Serre relations. Calling the $\hbar$ -deformation a “Yangian” is an abuse of terminology: Drinfeld (1985) defined Yangians for simple Lie algebras $\mathfrak{g}$ with non-degenerate Killing form and non-trivial root system; $\mathfrak{gl}_1 \otimes \Lambda_{\text{Muk}}$ has neither.

Heal. Drinfeld 1985 is more general than simple Lie algebras: the definition depends only on a finite-dimensional Lie algebra with an invariant symmetric non-degenerate bilinear form, which the Mukai pairing supplies. Chari–Pressley 1995 record the abelian-Heisenberg case $Y_h(\mathfrak{h}_n)$ with arbitrary symmetric form. The word *Yangian* is therefore earned; *abelian* clarifies that the Lie bracket is trivial (the defining relation has no quadratic Y^2 term, only Heisenberg commutators), distinct from the non-abelian Yangian $Y_h(\widehat{\mathfrak{g}})$ of a simple Kac–Moody $\mathfrak{g}$. The R -matrix R_{K3} and the coproduct Δ_z both deform non-trivially in $\hbar$, so the Yangian is non-trivial despite the abelian Lie input.

Cycle 2: Is it really abelian?

Attack. The Hopf algebra has spectral coproduct Δ_z with z -dependent cross-terms starting at spin $s = 2$. The opposite coproduct $\Delta_z^{\text{op}} \neq \Delta_z$ unless $z = 0$. This is manifestly non-abelian at the Hopf level. Calling $Y_h(\mathfrak{g}_{K3})$ “abelian” is misleading.

Heal. The word *abelian* refers to the Lie-algebra bracket on a single Miki direction, not to the Hopf structure. On direction i , the Heisenberg commutator $[J_{i,m}, J_{i,n}] = \varepsilon_i \hbar m \delta_{m+n,0}$ is central, so the derived subalgebra is one-dimensional. The Hopf non-cocommutativity at $z \neq 0$ is a feature of the $\hbar$ -deformation, not a defect of the abelianness. Remark 1.7 in the body formalises this: *abelian-at-Lie* / *non-abelian-at-Hopf* is a scope declaration, not an inconsistency.

Cycle 3: CoHA $\neq$ positive half?

Attack. Schiffmann–Vasserot 2012 identify $\text{CoHA}(\mathbb{C}^3)$ with $\mathcal{W}_{1+\infty}$ at generic parameters; this is the full Yangian double, not the positive half. By analogy, $\text{CoHA}(K3)$ might be the full K3 Yangian $Y_h(\mathfrak{g}_{K3})$, not just $Y_h^+(\mathfrak{g}_{K3})$.

Heal. The cache fact is definitive: $\text{CoHA}(\mathbb{C}^3) = Y^+(\widehat{\mathfrak{gl}}_1)$, the positive half. $\mathcal{W}_{1+\infty}$ is the full Yangian double, obtained from Y^+ by Drinfeld doubling; CoHA gives only the positive half because the pushforward-in-short-exact-sequences product is a nilpotent, upper-triangular construction (sheaf extensions go in one direction). The same logic applies at K3: Schiffmann–Vasserot 2017 supplies $\text{CoHA}(K3) = U(Y_h^+(\mathfrak{g}_{K3}))$, not the full $Y_h(\mathfrak{g}_{K3})$. Recovery of $Y_h(\mathfrak{g}_{K3})$ requires the Drinfeld double (23), which is additional structure beyond the CoHA itself.

Cycle 4: MO stable envelope fails globally

Attack. The Maulik–Okounkov stable envelope requires a $(\mathbb{C}^*)^2$ -action; a generic K3 surface has $\dim \text{Aut}(K3)^\circ = 0$ (no non-trivial continuous automorphism). So the MO construction cannot exist globally on K3 moduli, falsifying Proposition 5.2.

Heal. Proposition 5.2 is scope-limited to ADE/Kummer charts, where the McKay correspondence supplies a local toric torus $T_S = (\mathbb{C}^*)^2$ on the analytic chart (the resolved singularity $\widehat{\mathbb{C}^2/\Gamma}$ has a torus action inherited from $\mathbb{C}^2$). Globally at generic K3 the MO machinery fails, but the three routes agree on the overlap where all three are defined. The statement `lem:mo-bypass-local-to-global` of `chapters/examples/k3_yangian_chapter.tex` makes this explicit: $\dim \text{Aut}(S \times E)^\circ = 1 < 2$ generically, since $\text{Aut}(E)^\circ \simeq E$ (abelian variety, no $\mathbb{G}_m$ -subgroup), so even after the elliptic factor the torus rank is 1.

Cycle 5: Physical hCS avatar is speculative

Attack. Costello–Gaiotto’s 4d Chern–Simons boundary Yangian is a theorem (Costello 2013, Costello–Witten–Yamazaki 2018); 5d hCS is partially constructed (Costello 2016); hCS on $S \times C$ is speculative. Theorem 6.1 is therefore a conjecture, not a theorem.

Heal. The boundary-algebra identification is conjectural at the level of the physical theory. What is rigorous is the E_1 -chiral algebra $Y_h(\mathfrak{g}_{K3})$ constructed in §1; what is conjectural is the physical identification with an hCS theory on $S \times C$. The rectification: Theorem 6.1 is labelled “Conjectural” in the body, and Proposition 6.2 is stated as conditional on that construction being valid. The mathematical content (Drinfeld Yangian axioms, RTT relations, Hopf structure, CoHA identification at the generator-relation level, MO agreement at ADE charts) is independent of the hCS interpretation.

Cycle 6: Mukai indefinite signature obstruction (confirmatory)

Attack. The Mukai pairing has signature (4, 20): 20 directions of “negative-level” Heisenberg. Negative-level Kac–Moody algebras often have degenerate or ill-defined representations (non-unitary at the unitary level, no positive-definite Shapovalov form). This could obstruct the Yangian construction or spoil the R -matrix unitarity.

Heal. Proposition `prop:mukai-indefinite-yangian` of `chapters/examples/k3_yangian_chapter.tex` proves that signature-(4, 20) is not an obstruction: each of the 24 directions is a rank-1 Heisenberg at level $\varepsilon_i = \pm 1$, and the Yangian construction depends only on $\varepsilon_i \neq 0$, not on its sign. The cross-sector commutator vanishes, so the algebra factorises as $Y(H_+) \otimes Y(H_-)$ with

$H_{\pm}$ the positive/negative Mukai sectors. The R -matrix is diagonal with each entry unitary: $(u - h)/(u + h) \cdot (-u - h)/(-u + h) = 1$ for each factor. The Fock-space inner product has signature $(4, 20)$, matching the Mukai pairing; positive-definite BPS representations exist for Donaldson–Thomas subspaces (sector factorisation $Y(H_+) \otimes Y(H_-)$). The signature enters through the R -matrix factorisation $g_{K3} = g_+ \cdot g_-$ of (U4), not through a Lie-bracket pathology.

Cycle 7: Is $\text{CoHA}(K3) = U(Y_h^+(\mathfrak{g}_{K3}))$ a theorem or a conjecture?

Attack. The 2012 Schiffmann–Vasserot theorem for $\text{CoHA}(\mathbb{C}^3)$ is established; the 2017 K3 extension ([arXiv:1705.07488](#)) is cited as Theorem 4.1 but the K3 case involves subtleties absent from the local surface case: the moduli stack $M_{K3}^{(v)}$ of sheaves on K3 is not a global quotient, the structure constants of the CoHA involve virtual fundamental classes requiring Behrend functions, and the identification with $U(Y_h^+(\mathfrak{g}_{K3}))$ appeals to a positive-half shuffle presentation that has not been proved in print for indefinite-signature lattices.

Heal. The primary literature is finer-grained than a single theorem: Schiffmann–Vasserot 2017 proves the CoHA of K3 is a graded Hopf algebra with primitive part Λ_{Muk} and that the positive-half shuffle presentation holds at the level of generating functions; the full identification $\text{CoHA}(K3) \cong U(Y_h^+(\mathfrak{g}_{K3}))$ at the generator-relation level requires the Yangian $Y(Y_h^+(\mathfrak{g}_{K3}))$ to be generated by its Mukai primitives, which follows from the abelian-at-Lie triangular decomposition of Proposition 1.6. The remaining subtlety — whether the structure constants of the CoHA match those of Y^+ over $\mathbb{Z}$ rather than over $\mathbb{Q}$ — is the content of Davison’s 2017 integrality extension ([arXiv:1712.06049](#)). Combined, the statement $\text{CoHA}(K3) \cong U(Y_h^+(\mathfrak{g}_{K3}))$ at the generator-relation level is a theorem modulo Davison integrality; at the level of the primitive part Λ_{Muk} it is a theorem of Schiffmann–Vasserot unconditionally. The scope of Theorem 4.1 in the body is the primitive-level + shuffle-level statement; the Z -integral enhancement is additional.

Cycle 8: Why does E_1 and not E_2 apply at K3?

Attack. K3 is a $d = 2$ CY surface; Φ_2 should send $D^b(\text{Coh}(K3))$ to an E_2 -chiral algebra (the factorisation algebra on $\mathbb{R}^2$), not E_1 . Calling $Y_h(\mathfrak{g}_{K3})$ an E_1 -chiral algebra contradicts the CY- A_2 functor-output scope.

Heal. The E_n scope depends on the chiral sector considered. The full E_2 -chiral image of $D^b(\text{Coh}(K3))$ on $\mathbb{R}^2$ is the Mukai–Heisenberg vertex algebra $V_{H_{\text{Muk}}}$, which supports genuine E_2 -chiral factorisation. The abelian K3 Yangian $Y_h(\mathfrak{g}_{K3})$ is the mode algebra of $V_{H_{\text{Muk}}}$ on a formal disk, an E_1 -chiral (associative-algebra) extraction; the E_2 structure is recovered by considering the full factorisation sheaf on $\mathbb{R}^2$ of which $Y_h(\mathfrak{g}_{K3})$ is the stalk. No contradiction: E_1 is the slice, E_2 the full stack. At $d \geq 3$ (i.e. $K3 \times E$) the ambient algebra A is only E_1 and the E_2 -chiral structure lives on $Z(\text{Rep}(A))$, not on A — a different cache fact (CLAUDE.md), distinct from the K3 case.

Verification summary

- *RTT equivalence.* Theorem 1.2: the diagonal R -matrix RTT relation reduces to the Heisenberg commutator on the diagonal. Independent verification by direct matrix computation at rank 3 (`compute/lib/k3_yangian_quantization.py`, 60 tests including explicit 27×27 YBE check at rank 3).

- *Rank-2 worked example* (Remark 1.3): entrywise diagonal compatibility; no non-trivial off-diagonal constraint.
- *Hopf structure*. Propositions 2.1–2.4: coassociativity, counit, antipode verified in `compute/lib/k3_rtt_ops.py` (52 tests).
- *Unitarity*. $R_{K3}(u) R_{K3}(-u) = I$ verified factor-by-factor.
- *CoHA identification*. Schiffmann–Vasserot 2017 [arXiv:1705.07488] gives $\text{CoHA}(K3) = U(Y_h^+(\mathfrak{g}_{K3}))$ at the primitive level + shuffle presentation; Davison 2017 [arXiv:1712.06049] provides integrality; the Drinfeld double extension restores $Y_h(\mathfrak{g}_{K3})$.
- *MO stable envelope at ADE*. Maulik–Okounkov 2012 [arXiv:1211.1287] applies at orbifold charts; BFN Coulomb branch at Kummer [arXiv:1604.03625] gives the third route.
- *Three R-matrix routes converge*. `compute/lib/mo_matrix_k3e.py` and `compute/lib/ade_yangian_level1.py` cross-check the agreement at ADE charts.
- *Indefinite signature*. Proposition `prop:mukai-indefinite-yangian` of the chapter, 60 tests in `mukai_indefinite_yangian.py`; sector decomposition, negative-level Heisenberg well-definedness, unitarity factor-by-factor.
- *Kunneth on κ_{cat}* . $\kappa_{\text{cat}}(K3 \times E) = \chi(\mathcal{O}_{K3}) \cdot \chi(\mathcal{O}_E) = 2 \cdot 0 = 0$; cache fact, matches the body §7.
- *E-scope*. $Y_h(\mathfrak{g}_{K3})$ is E_1 -chiral as a mode algebra; the underlying Mukai-Heisenberg vertex algebra $V_{H_{\text{Muk}}}$ is E_2 -chiral. Consistent with CY- A_2 output-scope and with cache-fact “ $d \geq 3$: A is E_1 ” applied one-step higher.